CM0832 - MT5001 Elements of Set Theory
2.1 Ordered Pairs

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Outline

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Preliminaries

Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Introduction

Starting...

> *"We begin our program of developing set theory as a foundation for mathematics by showing how various general mathematical concepts, such as relations, functions, and orderings can be represented by sets." (p. 17)*

Outline

Ordered Pairs

Definition [2.]1.1

Let a and b be sets. We define the **ordered pair** of a and b, denoted (a, b), where a is the **first coordinate** and b is the **second coordinate**, using Kuratowski (1921)'s definition, that is,

$$(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}.$$

Ordered Pairs

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Question

Is (a, b) a set?

Ordered Pairs

Example

We show that $(\emptyset, \{\emptyset\}) \dot{\neq} (\{\emptyset\}, \emptyset)$.

$$(\emptyset, \{\emptyset\}) \stackrel{\text{def}}{=} \{\{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \dot{\neq} \{\{\{\emptyset\}\}, \{\{\emptyset\}, \emptyset\}\} \stackrel{\text{def}}{=} (\{\emptyset\}, \emptyset).$$

Ordered Pairs

Example

Let a be a set. Then

$$\begin{aligned}(a, a) &\stackrel{\text{def}}{=} \{\{a\}, \{a, a\}\} \\ &\doteq \{\{a\}, \{a\}\} \\ &\doteq \{\{a\}\}.\end{aligned}$$

Ordered Pairs

Exercise [2.]1.1

Let a and b sets.

(a) Prove that $(a, b) \in \mathcal{P}(\mathcal{P}(\{a, b\}))$.

(b) Prove that $a, b \in \bigcup(a, b)$.

(c) Prove that if $a \in A$ and $b \in A$, then $(a, b) \in \mathcal{P}(\mathcal{P}(A))$.

Ordered Pairs

Theorem [2.]1.2

Let a, a', b, b' be sets. Then,

$$(a, b) \doteq (a', b') \quad \text{if and only if} \quad a \doteq a' \text{ and } b \doteq b'.$$

Ordered Pairs

Theorem [2.]1.2

Let a, a', b, b' be sets. Then,

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Exercise

To give a different definition of ordered pair. Note that your definition must be a set and it must satisfies Theorem [2.]1.2.

Ordered Pairs

Definitions (p. 18)

Let a, b, c and d sets. Then

$$(a, b, c) \stackrel{\text{def}}{=} ((a, b), c) \qquad \textbf{(ordered triples)}$$

$$(a, b, c, d) \stackrel{\text{def}}{=} ((a, b, c), d) \qquad \textbf{(ordered quadruples)}$$

$$(a) \stackrel{\text{def}}{=} a \qquad \textbf{(one-tuples)}$$

Outline

References

Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 4).

Casimir Kuratowski (1921). Sur la notion de l'ordre dans la Théorie des Ensembles. Fundamenta Mathematicae 2, pp. 161–171. DOI: 10.4064/fm-2-1-161-171 (cit. on pp. 7, 8).